

Sleep, Doomscrolling and the Night

How late-night scrolling, negative news, and protective routines relate to sleep quality across 1,000 respondents

47.6%

DOOMSCROLLERS

+10.6 min

LATENCY GAP

11 / 204

HEAVY-SCROLLER EXCEPTIONS

Executive summary

The answer first: the strongest signal is the bedtime-disruption chain

Doomscrolling is consistently associated with a harder night in this dataset. Respondents labeled doomscrollers take longer to fall asleep, wake more often, carry more weekly sleep debt, and sleep slightly less. Bedtime screen time is the clearest continuous marker.

The relationship is not destiny. A small group of heavy scrollers still reports good sleep. Their profile shifts the useful question from who scrolls to what surrounds the scrolling.

10.6 min

LONGER LATENCY

1.5 h

MORE WEEKLY DEBT

15 min

LESS SLEEP NIGHTLY

- Doomscrolling plus negative-news consumption aligns with the highest anxiety, stress, and fatigue averages.
- Reading, meditation or journaling, a phone outside the bedroom, and regular exercise show more protective patterns than night mode alone.
- Behavioral variables predict sleep-quality category more strongly than most demographics.
- The data appear synthetic or simulation-assisted; findings are descriptive, non-causal, and not population prevalence.

CENTRAL THREAD

Exposure links to latency; latency, wakeups, and short sleep accumulate into debt and fatigue.

HERO FINDING

More scrolling shifts the night toward poorer sleep

Nightly doomscroll load combines session count with average duration. Poor sleepers sit farther to the right, while the overlap makes clear that exposure raises risk without determining the outcome.

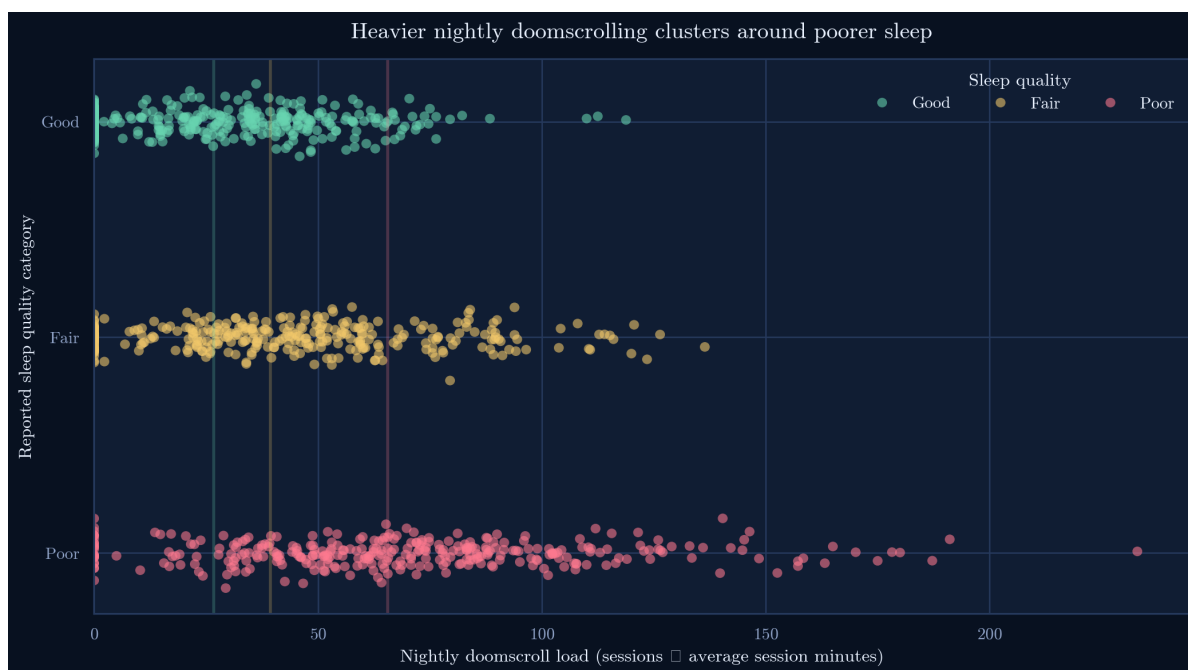


Figure 1. Nightly doomscroll load by reported sleep-quality category.

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Poor sleepers carry the highest median nightly doomscroll load; good sleepers cluster lower, with meaningful overlap.

Core relationship

Q1-2 / Sleep quality, latency, wakeups, duration, and debt

Across five outcomes, the doomscroller label points in the same direction: longer latency, more wakeups, more debt, shorter sleep, and greater fatigue. Shared causes such as stress, habit intensity, or chronotype may explain part of every gap.



Figure 2. Mean sleep outcomes for doomscrollers and non-doomscrollers.

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Doomscrollers average 10.6 extra minutes of latency and 1.5 additional hours of weekly debt, plus more wakeups and fatigue.

Dose response

Bedtime exposure matters beyond the binary label

Quartiles reveal a graded pattern: as bedtime exposure rises, latency and weekly sleep debt increase. Wakeups rise and sleep duration declines, suggesting the binary label compresses a richer relationship.

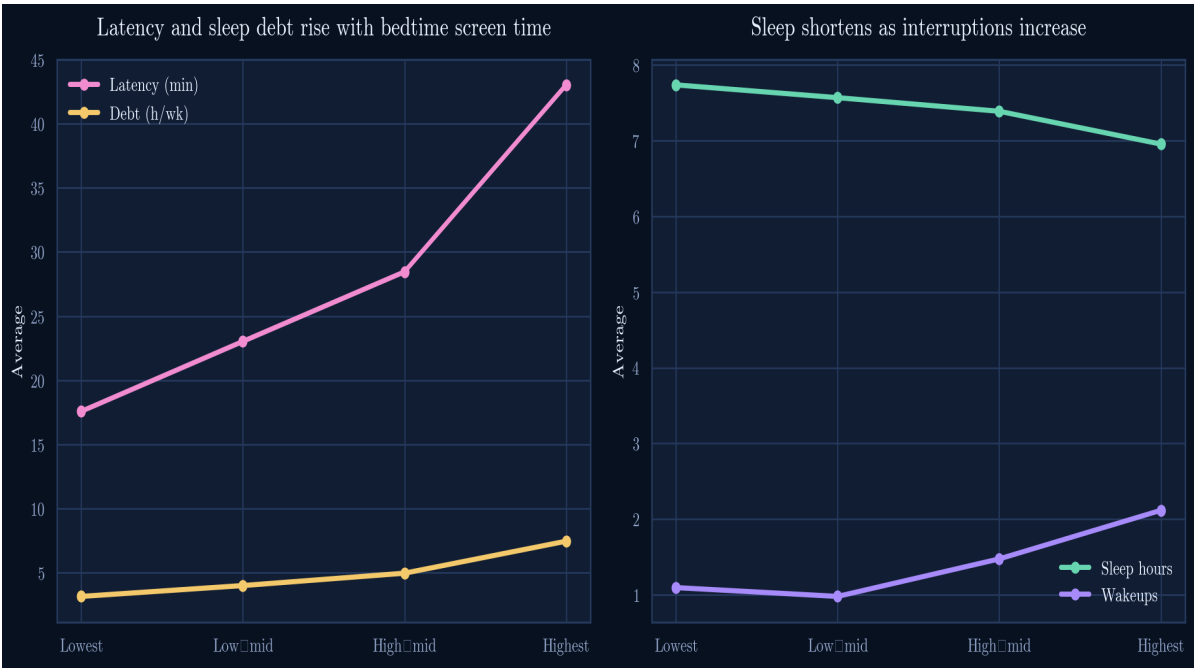


Figure 3. Average outcomes across quartiles of bedtime screen time.

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Latency and debt climb across exposure quartiles while sleep duration declines and wakeups rise.

Mental health angle

Q3-4 / Anxiety, stress, fatigue, and negative-news consumption

Doomscrolling involves content and emotional activation as well as minutes. When negative-news consumption accompanies doomscrolling, average anxiety, stress, and fatigue are highest. A feedback loop is plausible, but direction cannot be established.

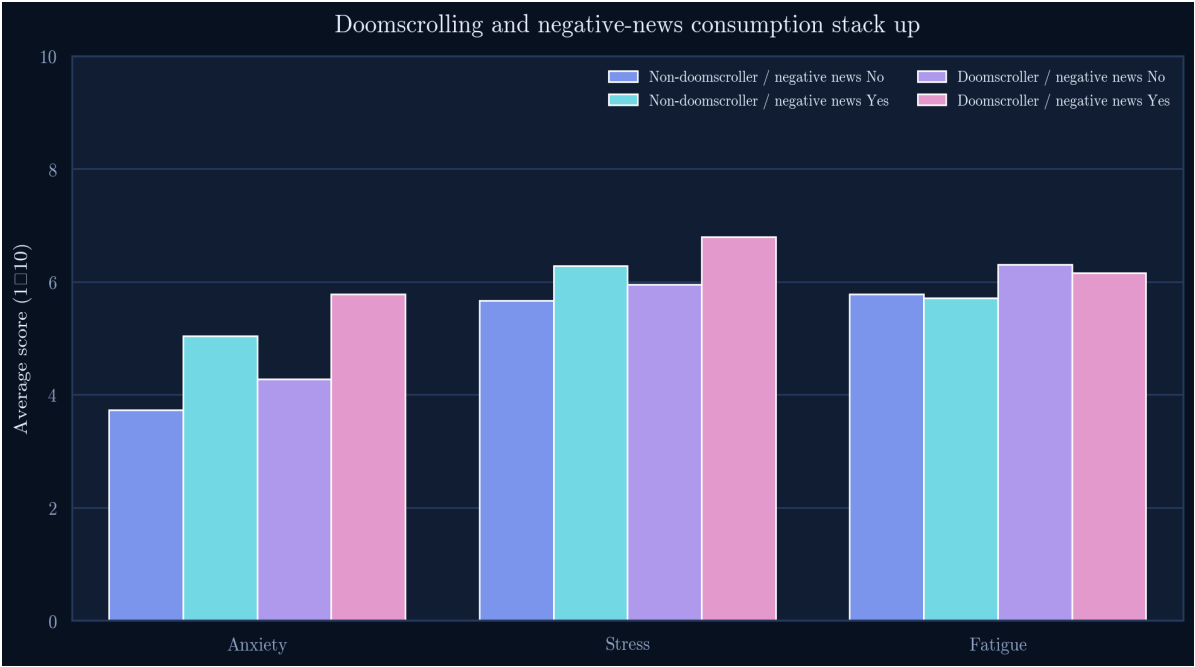


Figure 4. Average anxiety, stress, and fatigue by doomscrolling and negative-news status.

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The dual-exposure group shows the heaviest mental-wellbeing burden.

INTERPRET CAREFULLY

The dataset cannot tell whether distress drives scrolling, scrolling drives distress, or both.

Protective habits

Q5-6 / What appears to soften the pattern?

The most useful signals are structural: what a respondent does before bed, where the phone sleeps, and whether daytime movement is protected. Night mode is convenient, but its profile is weaker than a genuinely different bedtime routine.

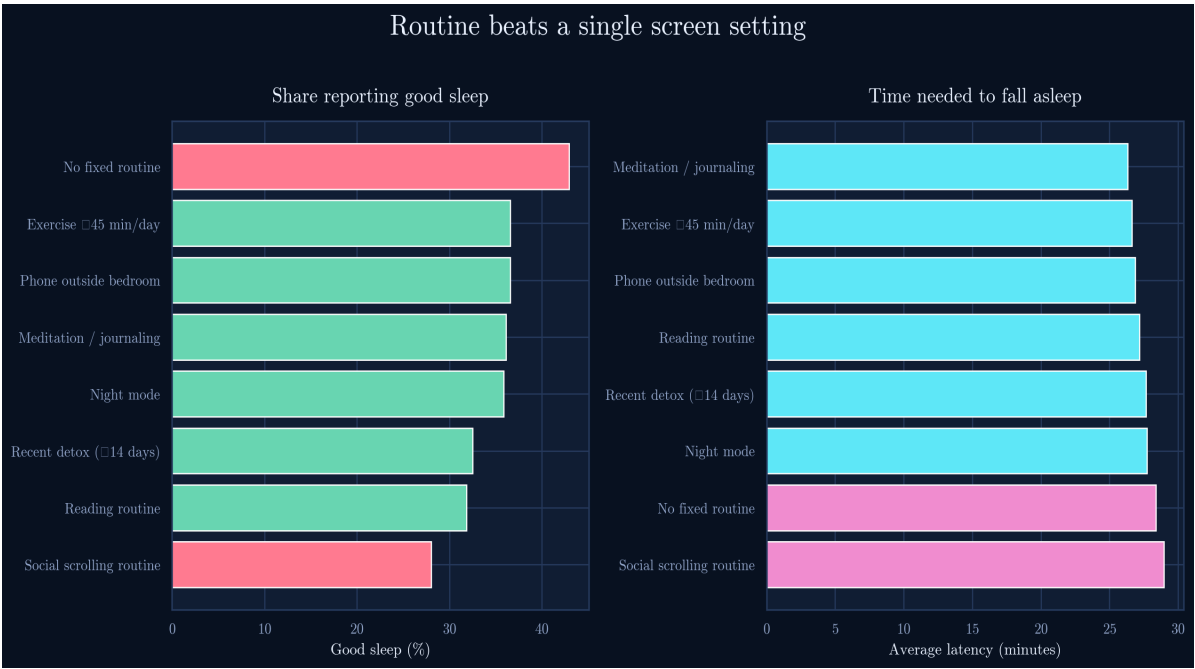


Figure 5. Good-sleep share and sleep latency for selected habits and routines.

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Reading and meditation or journaling stand out; social scrolling performs worst, and night mode is not a substitute for routine change.

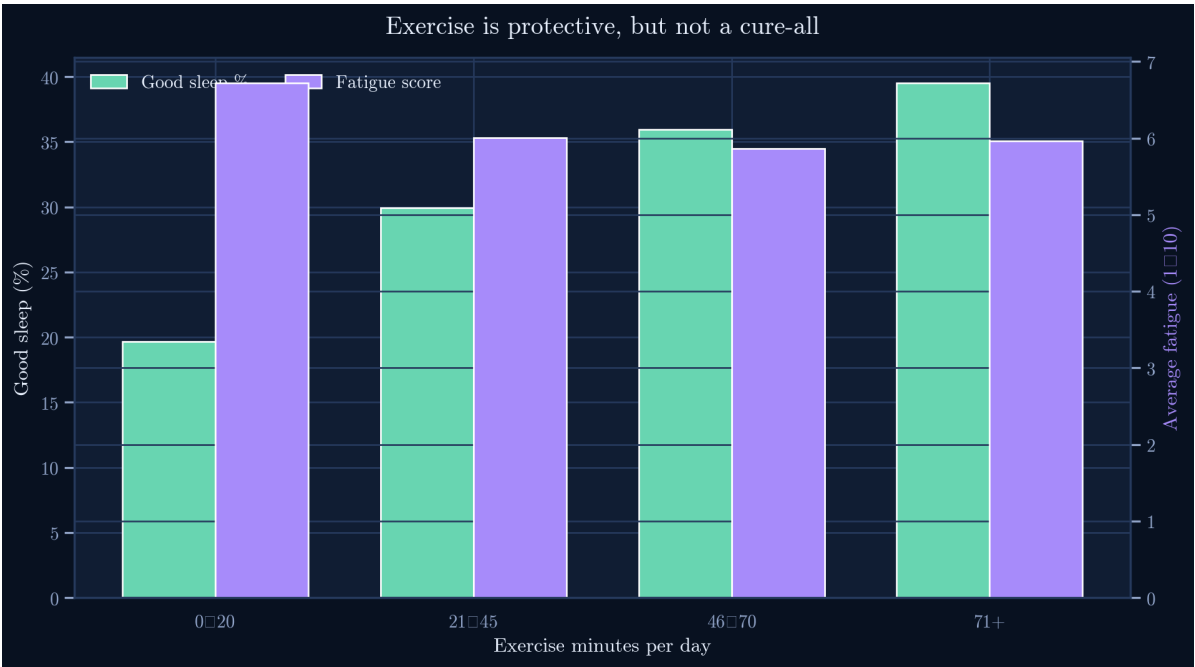


Figure 6. Good-sleep share and daytime fatigue across exercise bands.

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Exercise looks modestly protective, but the pattern is not perfectly monotonic and is not a cure-all.

Demographic context

Q7-8 / Age buckets, occupation, country, and descriptive gender context

Age buckets replace raw age where they clarify the story: Teens (15-19), 20s (20-29), and 30s+. Age-occupation cells reveal pockets of elevated doomscrolling. Country rates vary across samples of very different sizes, so these are segmentation clues rather than rankings.

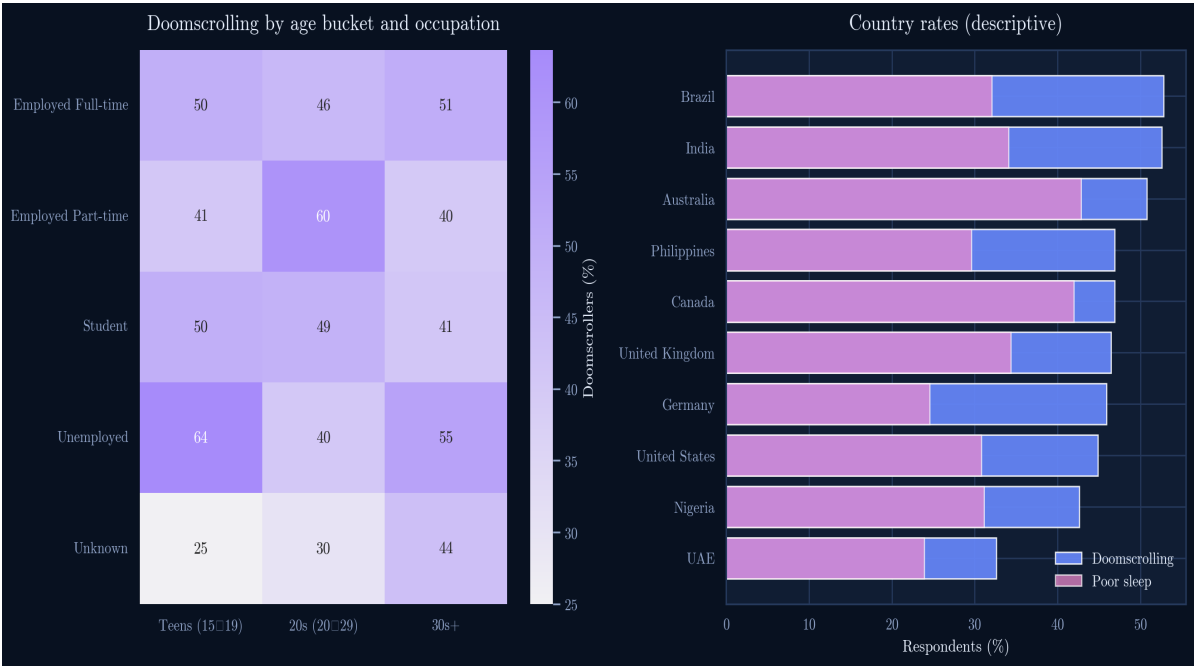


Figure 7. Doomscrolling by age bucket and occupation, plus country-level doomscrolling and poor-sleep shares.

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Younger and student pockets are often elevated, but small cells and uneven country samples prevent definitive comparisons.

GENDER

Gender differences are modest and descriptive only; behavior remains the more actionable signal.

The Exceptions

Counter-narrative / Heavy scrollers who still report good sleep

Heavy doomscrollers are self-identified doomscrollers at or above 66 minutes of bedtime screen time. There are 204 such respondents, yet only 11 report good sleep. The group is small enough to demand caution, but distinct enough to challenge a simplistic conclusion.



Figure 8. Standardized differences between good-sleep exceptions and other heavy scrollers.

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The exceptions realize more sleep and less debt or latency, with restorative routines appearing more often.

COUNTER-NARRATIVE

Exposure raises risk; routine and realized sleep appear to shape whether that risk becomes an outcome.

Personas

Transparent combinations for product and behavior design - not diagnoses

n=204

THE NIGHT SCROLLER

High bedtime exposure. First lever: environmental friction and a hard stop cue.

n=51

THE ANXIOUS NEWS SEEKER

High negative-news load and distress. First lever: a content boundary and decompression ritual.

n=130

THE DISCIPLINED SLEEPER

Lower exposure with reading or reflection. Opportunity: protect and reinforce the routine.

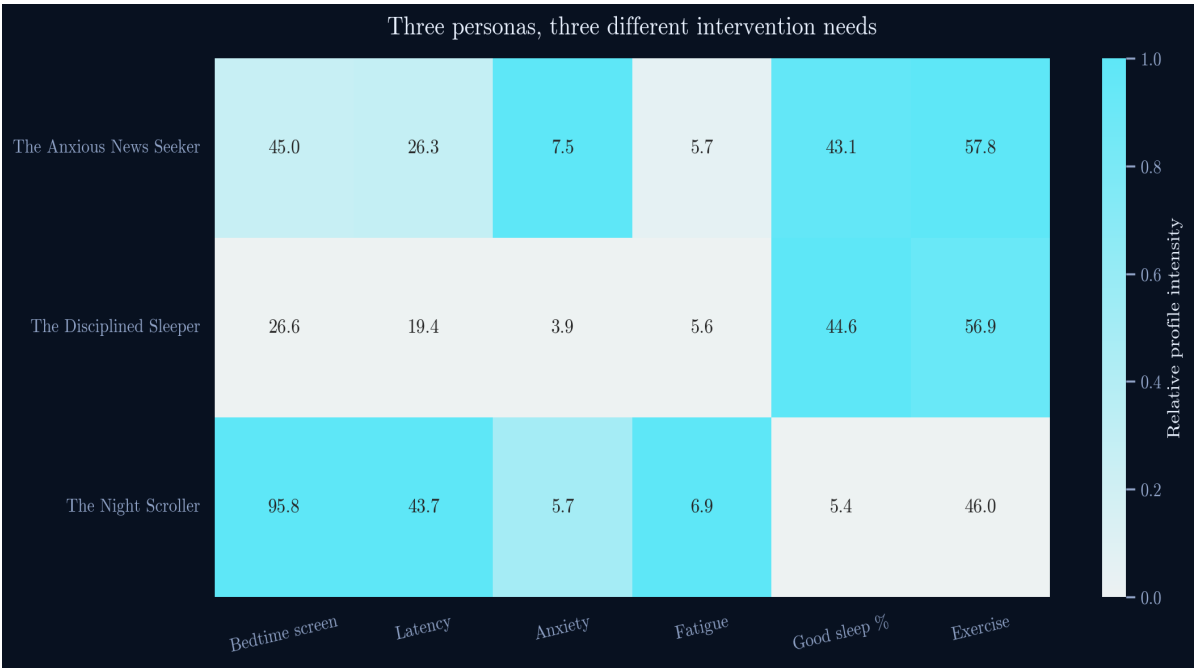


Figure 9. Relative persona profiles with observed metric values.

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Different mechanisms imply different interventions: friction, content boundaries, or routine reinforcement.

Synthesis

Q9 / Correlation structure and predictive ranking

The correlation structure resembles a bedtime-disruption chain. Exposure relates to latency; latency, wakeups, and shorter sleep accumulate into debt and fatigue. This coherence also heightens the synthetic-data caveat.

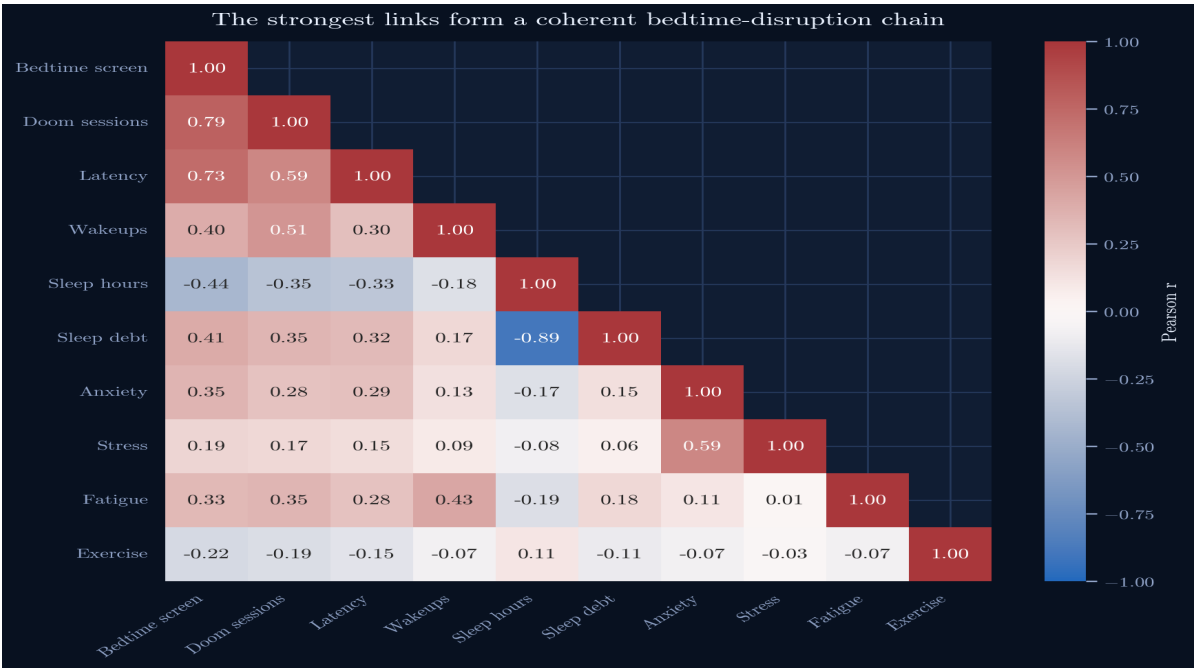


Figure 10. Pearson correlations among selected behavioral, sleep, and wellbeing measures.

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Bedtime screen time, doomscroll sessions, and latency form the strongest exposure cluster; sleep duration moves opposite sleep debt.

Predictive view

The behavior label leads; the night’s mechanics follow

A random-forest classifier evaluated with five-fold cross-validation reaches 60.3% balanced accuracy against a 34.0% majority-class baseline. The outcome-adjacent sleep-quality score is excluded to reduce leakage.

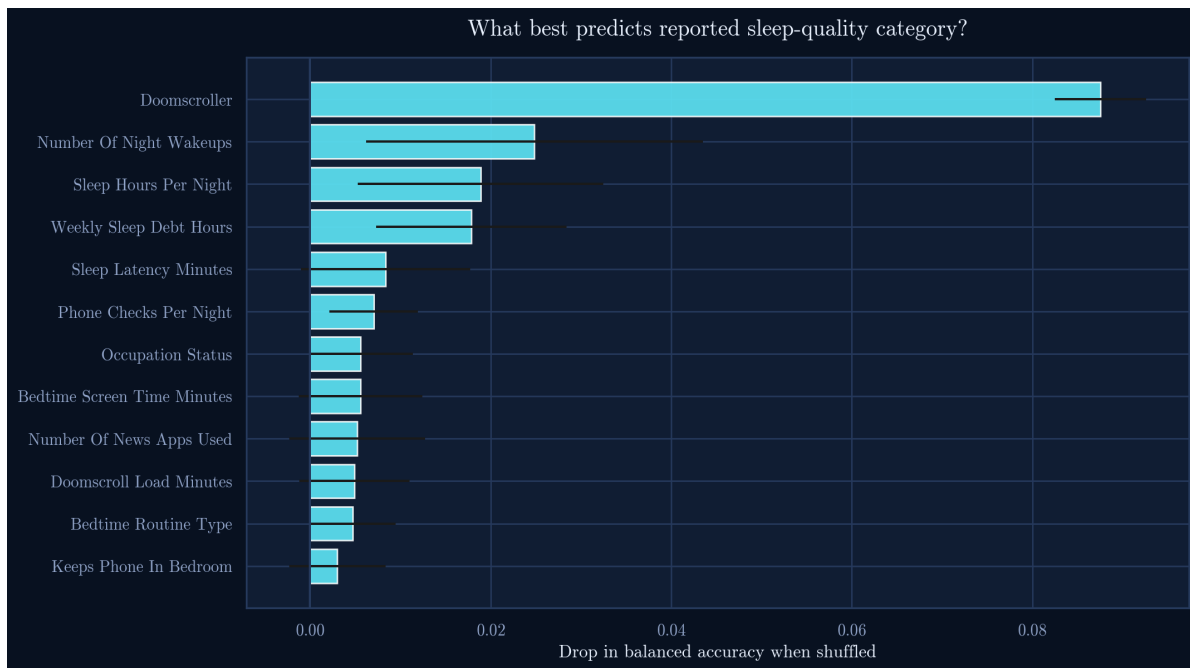


Figure 11. Cross-validated permutation importance for predicting sleep-quality category.

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The doomscribler label leads, followed by wakeups, sleep duration, sleep debt, and latency. Prediction does not identify causes.

Conclusion

Practical takeaways and limits

The defensible conclusion is not that every minute of scrolling destroys sleep. A repeatable pattern - longer exposure, more checking, delayed sleep, more wakeups, and accumulated debt - appears throughout the data. Action should target the chain rather than moralize the behavior.

Practical takeaways

- Create friction before bedtime: charge the phone outside the bedroom or beyond arm's reach.
- Replace, do not merely remove: use reading, meditation, or journaling for the final 20-30 minutes.
- Treat negative news as a separate lever: schedule a news window and disable late alerts.
- Protect daytime movement without expecting exercise to offset severe nighttime disruption.
- Track latency, wakeups, and weekly debt - not only total screen time.
- Test by persona: friction for exposure, content boundaries for distress, maintenance cues for disciplined sleepers.

BEST NEXT TEST

A two-week phone-outside-bedroom experiment with a fixed replacement routine and daily latency tracking.

Limitations

- The dataset appears synthetic or simulation-assisted: target classes are unusually balanced, relationships are unusually clean, and valid sleep-quality scores are concentrated at 5/5.
- The design is cross-sectional and self-reported; temporal direction and causal effects cannot be established.

- Median imputation and an Unknown category preserve all 1,000 records but can attenuate associations.
- Some subgroup cells - especially the good-sleep heavy-scroller exception group - are small and exploratory.
- Country and gender comparisons are descriptive only; unequal samples and unmeasured context prevent broad generalization.

Method in brief

The workflow validates IDs, dtypes, missingness, ranges, and formulaic patterns; creates age buckets; compares groups and quartiles; defines exceptions and personas; visualizes correlations; and evaluates a random forest with stratified five-fold cross-validation. The executed notebook remains viewable on the methodology page.

Synthetic-data audit

SANITY-CHECK SIGNAL	OBSERVATION	REPORTING CONSEQUENCE
Target balance	Only ten rows separate the three sleep-quality classes.	Do not treat class shares as population prevalence.
Ceiling effect	More than 86% of valid quality scores are 5/5.	Use categories cautiously; retain the caveat in every synthesis.
Orderly dependence	Sleep duration and weekly debt correlate at about -0.89.	Keep effect sizes inside this file; avoid causal or external claims.
Formula scan	No exact one-variable arithmetic formula reproduces the target.	Synthetic appearance is a limitation, not proof of a generator.

REPRODUCIBILITY

The PDF is the sole report export. The executed notebook remains embedded as a read-only methodology artifact on the website; chart interactions are powered by Plotly from a generated JSON specification.

Interpretation boundary

OBSERVATION	PREDICTION	SEGMENTATION
Pattern, not proof A repeated association is a reason to test a bedtime intervention. It is not evidence that one behavior caused an individual outcome.	Rank, not mechanism Feature importance shows which variables help classify sleep quality in this file. It does not reveal a biological or behavioral mechanism.	Context, not ranking Age, occupation, country, and gender describe where patterns concentrate. Unequal samples prevent cultural or demographic league tables.

THE USEFUL QUESTION

What can change around the scrolling?

Start with friction, content boundaries, and a replacement routine; then watch latency, wakeups, and weekly debt.